



App Name: Parent Post

Team Name: Mindful Mates

Team number: 52

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Introduction

Effective communication between teachers and parents plays an indispensable role in supporting learners' academic success, personal development and overall well-being. However, in many schools, communication gaps between teachers and parents often lead to missed opportunities to address students' learning needs, unmonitored behavior and misunderstandings that can negatively impact a child's educational experience. Many parents remain unaware of their children's academic progress, attendance patterns and/or behavioral issues due to a lack of accessible and timely communication channels.

To address this challenge, our team developed Parent Post, a web-based application using Oracle Apex, to strengthen collaboration between teachers and parents in both primary and high schools. The platform enables teachers to share learners' academic results, attendance records and behavioral reports directly with parents, ensuring transparency and continuous engagement.

By fostering consistent, clear and structured communication, Parent Post promotes a stronger partnership between parents and teachers, resulting in improved academic performance, timely interventions and enhanced learner wellbeing. This application also supports isolation access, ensuring data privacy and individualized functionality.

1. Requirements and their limitations

1.1 Functional Requirements

The Parent Post web application is designed to enhance communication between parents and teachers, and the following functional requirements were identified and implemented:

1.1.1 User authentication and role management

- The application supports secure login for multiple user roles: Teachers, Relatives, Clerks and Administrators.
- Each user can only access data relevant to their role and associate school, ensuring privacy and data separation between schools.

1.1.2 Learner Information Management

- The app maintains up-to-date records of learners including their personal details, grades, attendance and academic performance.

- Teachers can update attendance and results directly, while parents can only view their child's or children's information.

1.1.3 Teacher-Relative Communication

- A two-way private messaging system allows teachers to communicate directly about learners' performance, behavior and other concerns.
- Conversations are logged with timestamps and message read statuses to ensure accountability and continuity of communication.

1.1.4 Academic Results Management

- Teachers record learners results per subject and term.
- Relatives can view the marks, remarks and their subject-specific performance summaries through the relative dashboard.

1.1.5 Attendance and Behavior Tracking

- Teachers record daily attendance and behavioral observations when and where necessary.
- The data is automatically summarized into monthly reports accessible to both relatives and teachers.

1.1.6 Event Management

- The schools' clerks and teachers can publish events such as meetings, prize giving and other extracurricular activities.
- Relatives view these events published by the school to encourage greater participation in school activities.

1.1.7 School-Specific Data Segregation

- Each participating school has isolated data access.
- Teachers, clerks and relatives can only view and manage learners linked to their respective schools

1.1.8 Reports and Summaries

- The system automatically generates learner performance summaries and attendance reports.
- Teachers and relatives can view historical data for performance and intervention planning.

1.2 Non-Functional Requirements

1.2.1 Security and Privacy

- User authentication and role-based access control protect confidential learner data.
- Communication between users is stored securely to maintain privacy and traceability.

1.2.2 Usability

- The user interface is designed to be simple, user-friendly and accessible to both educators and parents with minimal technical knowledge.

1.2.3 Performance

- The system ensures fast response times when loading dashboards, reports and communication threads.

1.2.4 Scalability

- Parent Post can be expanded to include more schools and additional functionalities

1.2.5 Reliability

- The system maintains data integrity during updates and transactions to prevent loss of learner information

1.3 Limitations

1.3.1 Internet Dependence

- As a web-based platform, Parent Post requires stable internet connectivity. Users in rural or under-resourced areas may face difficulties accessing the platform consistently.

1.3.2 Limited Real-Time Notifications

- While internal messaging is supported, real-time alerts for new messages or updates are not yet fully implemented.

1.3.3 Mobile Responsiveness Constraints

- The system performs best on desktop browsers; some mobile screens may display minor alignment issues that require optimization.

1.3.4 Data Entry Reliability

- The accuracy of attendance, behavior and result records depends heavily on teachers' timely and correct data entry.

1.3.5 No Offline Mode

- Parent Post currently lacks offline capabilities; users cannot record or view data without an internet connection.

1.3.6 Limited Analytics and Reports

- Although the app generates summary reports, advanced analytics such as progress trends and performance predictions are planned for future updates.

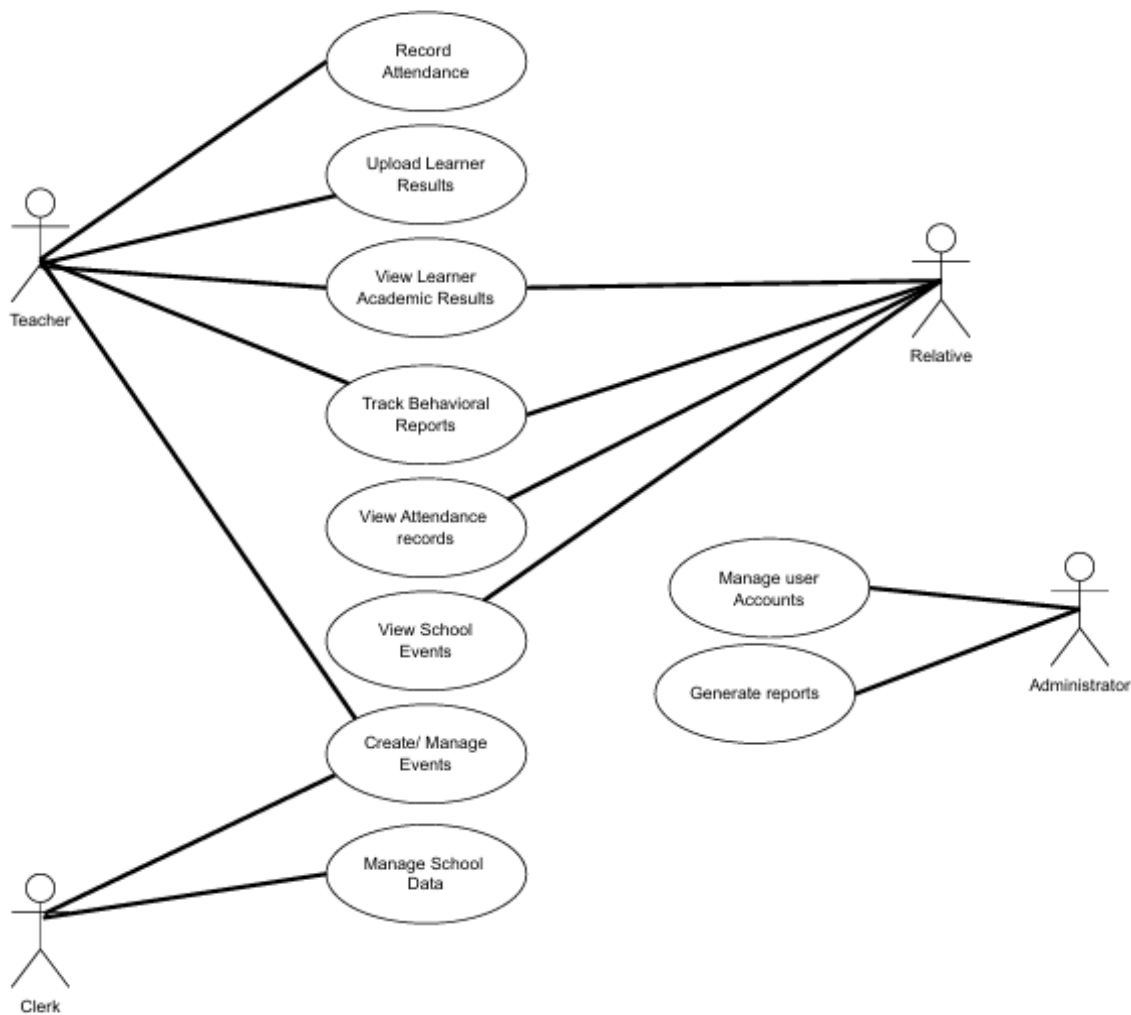
1.3.7 Role Expansion

- Future versions should include principal or district-level roles for broader school management and oversight capabilities.

2. Use-case Diagram and assessment of its quality

2.1 Use-case diagram

The use-case diagram below illustrates the main actors and their roles and how they interact with the system. It visually represents the functional requirements mentioned above because it demonstrates how different Teachers, Relatives, Clerks and administrators perform tasks through the system's interface.



2.2 Assessment of the use-case quality

The quality of the Parent Post use-case diagram was evaluated based on key modelling principles

- **Clarity-** The diagram is well-structured and uses standard UML notation, making it easy to understand the interactions between users and the system. Actors are clearly distinguished, and use cases are properly labelled.
- **Completeness –** It captures all essential system functionalities, including communication, result management, attendance tracking and event management. Each major user interaction is represented.

- Consistency – The diagram aligns with both the system requirements and database design. Each use-case corresponds to an implemented feature in the database schema and user interface.
- Correctness – The relationship between actors and use-cases correctly reflect system behavior.
- Simplicity – The diagram avoids unnecessary complexity by focusing only on relevant actors and core interactions. It provides a clear, high-level overview suitable for both technical and non-technical stakeholders.

3. Database Relations and an Assessment of Their Quality

3.1 Database tables and their Attributes

Persons (Pid(pk), Username, Fname, Lname, Cellno, Gender, Email, Address)

Schools (Sid(pk), Name, Address, Telephone, Semail)

Grades (Gid(pk), Gname, Sid(fk))

Learners (Lid(pk), Pid(fk), Sid(fk), Gid(fk))

Teachers (Tid(pk), Honorific, Position, Pid(fk), Sid(fk))

Relatives (Rid(pk), Pid(fk))

Clerks (Cid(pk), Honorific, Pid(fk), Sid(fk))

Subjects (Scid(pk), Sname, Sid(fk))

TeachersSubjects (TSID (pk), Year, Tid(fk), Scid(fk))

LearnersSubjects (LSID (pk), Year, Lid(fk), Scid(fk))

LearnersRelatives (LRid(pk), Relationship, Lid(fk), Rid(fk))

Attendance (Aid(fk), Day, Status, Tid(fk), Lid(fk))

Conversations (Conv_id(pk), Started_at, last_updated, Lid(fk), Tid(fk), Rid(fk))

Events (Eid(pk), Name, Description, Location, Eventdate, Time, Sid(fk))

Messages (Msg_id(pk), Sender_pid, Sender_role, Message_text, Created_at, Is_read, Conv_id(fk))

Results (Reid(pk), Mark, Term, Recorded_date, Remarks, Scid(fk), Sid(fk), Lid(fk))

TRInteractions (TRID(pk), Issue, Resolution, Date_, Place, Time_, Medium, Rid(fk), Tid(fk), Lid(fk))

3.2 Entity Relationships

- Persons to Teachers: this is a one-to-one relationship. We used the foreign constraint (PID) to ensure that each person is one teacher in the system, and now teachers only have one profile, there are no duplicates.
- Persons to Relatives: this is a one-to-one relationship. We used the foreign constraint (PID) here too to ensure that each person is one relative in the system, there are no duplicates.
- Persons to Clerks: this is a one-to-one relationship, and we used the foreign constraint (PID) to ensure that each person is one clerk and there are no duplicates.
- Persons to Learners: this is a one-to-one relationship. We used the foreign constraint (PID) to ensure that each person is one learner and there are no duplicates.
- Schools to Grades: this is a one-to-many relationship. Each school has many grades, and many grades belong to one school. This enforces the data isolation per school.
- Schools to Teachers: This is a one-to-many relationship. Each school has many teachers and teachers belong to only one school. This enforces the data isolation per school.
- Schools to Learners: This is a one-to-many relationship between schools and learners. Each school has many learners, and many learners belong to one school. This enforces the data isolation per school.
- Schools to Clerks: This is a one-to-one relationship between the schools and the clerks. Each school has only one clerk. This enforces the data isolation per school.
- Schools to Subjects: This is a one-to-many relationship between the schools and subjects. Each school has many subjects, and many subjects belong to one school. This enforces the data isolation per school.
- Grades to Learners: This is a one-to-many relationship between grades and learners. Each grade has many/multiple learners enrolled.
- Learners to Subjects: This is a many-to-many relationship between the learners and subjects. This is implemented through the LearnersSubjects junction table that allowed a learner to be enrolled in various subjects, vice versa.
- Teachers to Subjects: This is a many-to-many relationship between teachers and subjects. This is implemented through the TeachersSubjects junction table, linking teachers with the subjects they teach for a specific year.
- Learners to Relatives: This is a many-to-many relationship between learners and relatives. This is implemented or managed through the LearnersRelatives junction table, linking relatives with the learners they are parents/guardians for.
- Learners to Attendance: This is a one-to-many relationship. Each learner has multiple attendance records, one attendance record for each day.
- Learners/Teacher/Relatives to Conversations: This a one-to-many relationship – Conversations connect the two actors, which are the teachers and relatives, through learners in private communication threads.
- Conversations to Messages: This is a one-to-many relationship – Each conversation consists of multiple message entries, representing a full communication history.

- Schools to Events: This is a one-to-many relationship between schools and events – Each event is associated with a single school, allowing event announcements to be school-specific.
- Learners to Results: This is a many-to-many relationship – Each learner has multiple result records for different subjects, terms, and years.
- Teachers/Relatives/Learners to TRInteractions: This is a many-to-many relationship – The TRInteractions table captures specific issues or cases discussed between teachers and relatives about a particular learner.

3.3 Assessment of the database relations quality

The quality of the Parent post database was evaluated using key relational database design principles, which are normalization, referential integrity, data consistency and scalability.

- Normalization – The database is normalized up to Third Normal Form (3NF). Each table contains atomic attributes, and all non-key fields depend solely on the primary key. This eliminates redundancy and ensures efficient updates.
- Referential Integrity – All relationships are enforced through foreign key constraints, ensuring that data remains consistent and valid across tables. Deletion or modification of related records is restricted unless dependencies are handled.
- Data Consistency – Data is centrally managed through the Persons table, ensuring that personal details are consistent across roles.
- Scalability - The design supports adding more schools, learners and subjects without affecting performance or requiring structural redesign. Each school's data remains logically isolated.
- Flexibility – The use of junction tables makes the schema highly adaptable to changes in curriculum or relationships.
- Security and Privacy - Sensitive data such as personal details and messages are stored in separate, access-controlled tables. Each record is linked to a school identifier that is also a global item SID for isolation.
- Performance Efficiency – Indexed primary and foreign keys improve query response times for large datasets.
- Maintainability – The schema structure is modular and logically grouped, making it easier to maintain and extend with minimal disruption.

4. Description of the Main Functionality of the App

Parent Post is a web-based application designed to bridge the communication gap between teachers and parents or guardians. It provides them with a structured platform that allows them to monitor the attendance of a learner, academic performance and get monthly reports on the overall behavior of a learner. The system uses role-based dashboards to ensure that teachers,

relatives and clerks can have each other's responsibilities while maintaining data integrity and privacy through session variables G_SID (School ID), G_PID (Person ID), G_TID (Teacher ID) and G_RID (Relative ID) that are used in the procedure used in the login. Through a communication channel on the app and real-time updates, Parent Post promotes transparency, collaboration and overall well-being of a learner. This is the procedure that is used on the login:

```
create or replace PROCEDURE SET_LOGIN_ITEMS (  
    p_username IN VARCHAR2  
) AS  
    v_pid NUMBER;  
    v_tid NUMBER;  
    v_sid NUMBER;  
    v_rid NUMBER;  
BEGIN  
  
    SELECT pid  
        INTO v_pid  
        FROM persons  
        WHERE LOWER(username) = LOWER(p_username);  
  
    APEX_UTIL.SET_SESSION_STATE('G_PID', v_pid);  
  
    BEGIN  
        SELECT t.tid, t.sid  
            INTO v_tid, v_sid  
            FROM teachers t
```

```
JOIN persons p ON t.pid = p.pid  
WHERE LOWER(p.username) = LOWER(p_username);
```

```
APEX_UTIL.SET_SESSION_STATE('G_TID', v_tid);
```

```
APEX_UTIL.SET_SESSION_STATE('G_SID', v_sid);
```

```
EXCEPTION
```

```
WHEN NO_DATA_FOUND THEN
```

```
    APEX_UTIL.SET_SESSION_STATE('G_TID', NULL);
```

```
BEGIN
```

```
    SELECT r.rid
```

```
        INTO v_rid
```

```
        FROM relatives r
```

```
        JOIN persons p ON r.pid = p.pid
```

```
        WHERE LOWER(p.username) = LOWER(p_username);
```

```
    APEX_UTIL.SET_SESSION_STATE('G_RID', v_rid);
```

```
FOR rec IN (
```

```
    SELECT l.sid, l.lid
```

```
    FROM learnersrelatives lr
```

```
    JOIN learners l ON l.lid=lr.lid
```

```
    WHERE lr.rid = v_rid
```

```
) LOOP
```

```
    APEX_UTIL.SET_SESSION_STATE('G_SID', rec.sid);
```

```
END LOOP;
```

```
EXCEPTION
```

```
    WHEN NO_DATA_FOUND THEN
```

```
        APEX_UTIL.SET_SESSION_STATE('G_RID', NULL);
```

```
BEGIN
```

```
    SELECT c.sid
```

```
        INTO v_sid
```

```
        FROM clerks c
```

```
        JOIN persons p ON c.pid = p.pid
```

```
        WHERE LOWER(p.username) = LOWER(p_username);
```

```
    APEX_UTIL.SET_SESSION_STATE('G_SID', v_sid);
```

```
EXCEPTION
```

```
    WHEN NO_DATA_FOUND THEN
```

```
        APEX_UTIL.SET_SESSION_STATE('G_SID', NULL);
```

```
END;
```

```
END;
```

```
END;
```

END;

/

4.2 Teacher Dashboard

This dashboard appears only to teachers, and it provides them with a friendly, centralized interface to manage the tasks they need to achieve on the app and communications. Upon logging in and verifying, a teacher sees only the data that is related to them, which is their assigned school and learners. This dashboard has the following core functionalities:

- View learners' details – teachers are granted access to their learners' personal and academic information.
- Record attendance – the teachers mark the attendance register and justify absences if not reported on why they are absent, but this is implemented through conversations between the relative and the teacher.
- Record learner results – teachers are to enter and manage the marks for the subjects they teach.
- Send messages to relatives - the teachers initiate or reply to private conversations with learners' relatives or parents when they have something to discuss with their learners' relatives. This is the code for the process we created for this function:

```

DECLARE
    v_message_type VARCHAR2(20);
BEGIN
    IF LOWER(:P46_MESSAGE_TEXT) LIKE '%issue%' OR LOWER(:P46_MESSAGE_TEXT) LIKE
'%problem%' THEN
        v_message_type := 'ISSUE';
    ELSIF LOWER(:P46_MESSAGE_TEXT) LIKE '%resolved%' OR LOWER(:P46_MESSAGE_TEXT)
LIKE '%fixed%' THEN
        v_message_type := 'RESOLUTION';
    ELSE
        v_message_type := 'NORMAL';
    END IF;

    INSERT INTO messages (
        msg_id, conv_id, sender_pid, sender_role, message_text, message_type, created_at,
is_read
    ) VALUES (
        messages_seq.NEXTVAL,
        :P46_CONV_ID,
        :G_PID,
        :P71_SENDER_ROLE,
        :P46_MESSAGE_TEXT,
        v_message_type,
        SYSDATE,
        'N'
    );
END;
```

UPDATE conversations

SET last_updated = SYSDATE

WHERE conv_id = :P46_CONV_ID;

- Access reports - the teachers view monthly reports of the learners they teach only; these monthly reports are the behavioral reports and attendance reports. This is implemented via a wizard that has 3 steps. Step 1 requires the teacher to enter the learner ID (LID) of the learner for whom they want to view the monthly report. In step 2, the teacher is required to enter the month and the year of the report they want. We created a hidden page item LID in this step to carry the chosen LID from step 1. Then, moving to step 3, the LID, month and year are carried over to this step, where the report is displayed. We also used the bar graph for relatives to have a clearer picture of the reports. This is the code for the bar graph we did for attendance:

SELECT

'Days Present' AS LABEL,

TOTAL_PRESENT AS VALUE,

'#2ECC71' AS COLOR

FROM LEARNERS_ATTENDANCE

WHERE LID = :P45_LID

AND MONTH_NO = TO_NUMBER (:P45_MONTH)

UNION ALL

SELECT

'Days Absent' AS LABEL,

TOTAL_ABSENT AS VALUE,

'#E74C3C' as COLOR

FROM LEARNERS_ATTENDANCE

WHERE LID = :P45_LID

AND MONTH_NO = TO_NUMBER (:P45_MONTH)

This is the code for the bar graph of the behavioral report:

SELECT

'Total Issues' AS LABEL,

SUM(TOTAL_ISSUES) AS VALUE,

'#F39C12' AS COLOR

FROM MONTHLY_BEHAVIOR_REPORT

WHERE LID = :P45_LID

AND SUBSTR(REPORT_MONTH,6,2) = LPAD(:P45_MONTH,2,'0')

UNION ALL

SELECT

```

'Total Resolutions' AS LABEL,
SUM(TOTAL_RESOLUTIONS) AS VALUE,
'#8E44AD' AS COLOR
FROM MONTHLY_BEHAVIOR_REPORT
WHERE LID = :P45_LID
AND SUBSTR(REPORT_MONTH,6,2) = LPAD(:P45_MONTH,2,'0');

```

4.3 Relative dashboard

This relative dashboard authorizes the parents and guardians to stay informed about their children's academic results and behavioral progress, and each parent sees only their own child(ren)'s data. Here are the core functionalities the relative gains access to:

- View learners' academic results – relatives are authorized to see the term results of the subjects their child takes and the remarks the teachers wrote.
- Monitor attendance records – relatives view monthly and daily attendance logs for their children.
- View behavioral reports – relatives see the feedback from the teachers regarding their children.
- Messaging interface – relatives communicate privately with the teachers through the in-app chat feature on the app. This is the code for the process we created:

```

DECLARE
    v_message_type VARCHAR2(20);
BEGIN
    IF LOWER(:P55_MESSAGE_TEXT) LIKE '%issue%' OR LOWER(:P55_MESSAGE_TEXT) LIKE
'%problem%' then
        v_message_type := 'ISSUE';
    ELSIF LOWER(:P55_MESSAGE_TEXT) LIKE '%resolution%' OR
LOWER(:P55_MESSAGE_TEXT) LIKE '%resolved%' then
        v_message_type := 'RESOLUTION';
    ELSE
        v_message_type := 'NORMAL';
    END IF;

    INSERT INTO messages (
        msg_id, conv_id, sender_pid, sender_role, message_text, message_type, created_at,
is_read
    ) VALUES (
        messages_seq.NEXTVAL,
        :P55_CONV_ID,
        :G_PID,
        :P55_SENDER_ROLE,
        :P55_MESSAGE_TEXT,

```

```

        v_message_type,
        SYSDATE,
        'N'
    );

    UPDATE messages
    SET is_read = 'Y'
    WHERE msg_id = (
        SELECT MAX(msg_id)
        FROM messages
        WHERE conv_id = :P_CHAT_CONV_ID
        AND sender_role != :G_ROLE
    );
END;

```

```

    UPDATE conversations
    set last_updated = SYSDATE
    Where conv_id = :P55_CONV_ID;

```

- Access monthly reports – relatives view monthly reports of the learners they are guardians/parents to; these reports have the attendance summary and more details on the attendance of a learner and the behavioral reports of a learner. This is implemented via a wizard that has 3 steps. Step 1 is where a relative is required to enter the learner ID(LID) of the learner they want to view the monthly report for. In step 2, the relative is required to enter the month and the year of the report they want. In step 3, the LID, month and year are carried over to this step, where the monthly reports will be displayed. We also used the bar graph for relatives to have a clearer picture of the reports. This is the code for the bar graph we did for attendance:

```

SELECT

    'Days Present' AS LABEL,

    TOTAL_PRESENT AS VALUE,

    '#2ECC71' AS COLOR

FROM LEARNERS_ATTENDANCE

WHERE LID = :P70_LID

    AND MONTH_NO = TO_NUMBER(:P70_MONTH_NO)

UNION ALL

SELECT

```

```

'Days Absent' AS LABEL,
TOTAL_ABSENT AS VALUE,
'#E74C3C' as COLOR
FROM LEARNERS_ATTENDANCE
WHERE LID = :P70_LID
AND MONTH_NO = TO_NUMBER(:P70_MONTH_NO)

```

UNION ALL

SELECT

```

'Total Days' AS LABEL,
TOTAL_DAYS AS VALUE,
'#3498DB' as COLOR
FROM LEARNERS_ATTENDANCE
WHERE LID = :P70_LID
AND MONTH_NO = TO_NUMBER(:P70_MONTH_NO);

```

This is the code for the bar graph of the behavioral report:

SELECT

```

'Total Issues' AS LABEL,
SUM(TOTAL_ISSUES) AS VALUE,
'#F39C12' AS COLOR
FROM MONTHLY_BEHAVIOR_REPORT
WHERE LID = :P70_LID
AND SUBSTR(REPORT_MONTH,6,2) = LPAD(:P70_MONTH_NO,2,'0')

```

UNION ALL

SELECT

```

'Total Resolutions' AS LABEL,

```

```

SUM(TOTAL_RESOLUTIONS) AS VALUE,
'#8E44AD' AS COLOR
FROM MONTHLY_BEHAVIOR_REPORT
WHERE LID = :P70_LID
AND SUBSTR(REPORT_MONTH,6,2) = LPAD(:P70_MONTH_NO,2,'0');

```

4.4 Clerk dashboard

Clerks act as administrative support for the schools, and they are responsible for data consistency and updates. The following are the core functionalities:

- Register new users of their school – clerks can add new teachers, learners and relatives into the system of the school.
- Approve teacher assignments – link teachers to subjects
- Approve learner assignments – link learners to subjects
- Approve relatives' assignments – link relatives to learners who are registered in their school.

5. Unresolved Challenges and Recommendations

Although the Parent Posy application has fulfilled most of its primary objectives, several technical, functional and operational challenges were encountered during development. Some of these issues were partially resolved while others remain open to future improvement.

5.1 Unresolved Challenges

5.1.1 Limited Notification and Alert Mechanism

The current system allows messaging and communication between teachers and relatives; however, real-time notifications are not yet fully implemented. Users must manually refresh or navigate to the messages page to view updates. This limits responsiveness and reduces user management, particularly for busy relatives and/or teachers.

5.1.2 Limited Data Analytics and Insights

Although the application provides reports on attendance and performance, advanced analytics like trend visualizations or comparative analysis across schools are not yet available on the app. With this challenge, users cannot easily identify long-term learning trends or school-wide performance insights.

5.1.3 Incomplete Mobile Responsiveness

While Parent Post is designed to be web-based and partially mobile-friendly, some pages - particularly interactive reports and forms - don't fully adapt to smaller screens. Teachers and relatives using smartphones experience layout misalignment and scroll issues, and this reduces the usability for mobile users, especially the relatives who rely only on mobile devices. We can enhance the responsive theme in Oracle Apex using Universal Theme options and custom CSS adjustments. We can also conduct extensive cross-device testing and possibly develop a progressive web app version of Parent Post for seamless mobile performance.

5.1.4 Attendance and Behavior Data Input Efficiency

Recording attendance manually for each learner can be time-consuming, especially for classes that have many students. We can introduce batch attendance entry or bulk update forms, allowing teachers to mark multiple learners at once.

5.2 Recommendations for Future Development

- Enhanced Data Privacy and Security – Incorporate data encryption for sensitive fields such as contact numbers and email addresses and enforce strong password policies.
- Conduct more beta testing across all participating schools to collect more real-world feedback, fix usability issues and ensure robust performance under load.
- Continuous System Maintenance - Establish a maintenance plan that includes database backup, performance tuning and regular version upgrades in Oracle APEX.

6. Conclusion

The completion of the Parent Post project represents a significant achievement in strengthening collaboration and transparency between teachers and parents. The system was developed to address one of the major challenges within the education sector – poor communication between schools and families, which often leads to misunderstandings, missed learning support and reduced student performance. Through methodological analysis, database modelling and iterative development, our team successfully built a web-based application that connects teachers, relatives and clerks in a unified digital environment. This solution ensures that parents are kept informed and actively engaged in their children's academic journey, while the teachers are empowered with simplified tools for managing learner information.

The Parent Post App has managed to achieve several key outcomes:

- Established efficient, structured communication between parents and teachers through a secure two-way messaging system.
- Improved learner monitoring through integrated attendance, performance and behavior tracking.

- Ensured data isolation and privacy across schools using the G_SID session control mechanism.
- Enhanced academic transparency by enabling parents to access their children's progress reports and updates.
- Increased administrative efficiency through clerk-managed data validation and role-based dashboards.

Despite the limitations that were mentioned, Parent Post has proven its effectiveness and relevance in improving teacher-parent interaction. The platform encourages timely feedback, fosters parental participation and supports a culture of shared responsibility for learners' success. It demonstrates how technology can enhance educational collaboration and build trust between educators and the parents of the learners .

7. Annexures

Organizations

Blythwood institution

Address: Sokapase A/A, Ngqamakhwe, 4990, Eastern Cape

Representative of the school: Mr Majuru

Contact Details: tamajuru@gmail.com

Zwelihlangene Primary School

Address: Nomadamba A/A, Ncora, Cofimvaba

Representative: Mrs Bavuma

Contact Details: 083 506 0033

Nkwanca Public High School

Address: 5027 Yekani Street, Mlungisi, Queenstown, 5321, Komani

Amitek High School

Address: 231 Upper, Oxford St, Southernwood, East London, 5201

Sixishe Agricultural High School

Address: 1216 Mkefa, Ezibeleni, Komani